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----- Active space -----

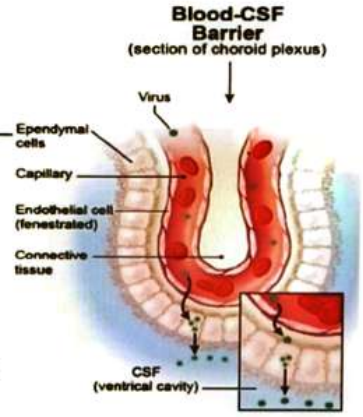
CSF, CBF AND BBB

Cerebrospinal Fluid (CSF)

00:01:50

General properties :

- Function : Lubrication & cushion.
- Site of production : Ependymal cells of choroid plexus.



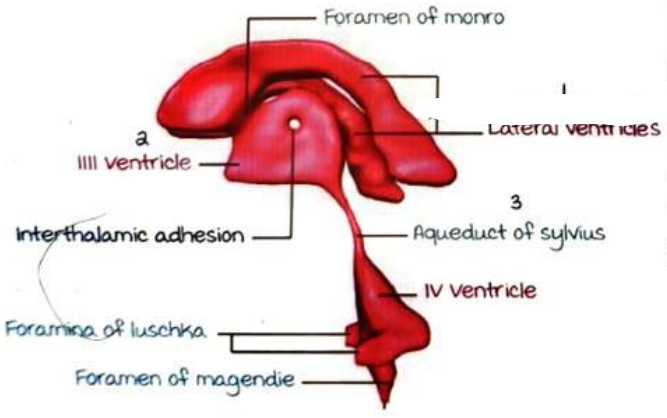
- Glial cell.
- Lines choroid plexus.
- Secretes CSF.

Parameters of CSF :

Parameters	Value
1. Volume	150 ml
2. Rate of production	550 ml/day
3. Turn over	3.7 times/day
4. CSF pressure - Normal. - Average.	70–180 mm H_2O. 112 mm H_2O. (Filtration = absorption).

Regulation : At the level of absorption (↓ absorption → ↑ CSF volume → ↑ CSF pressure).

Circulation :



4. Subarachnoid space

5. Arachnoid villi

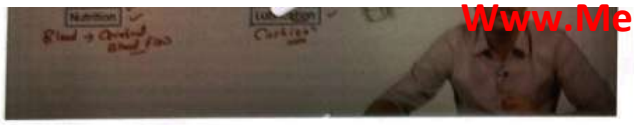
6. Venous sinuses : • Absorbs CSF.

CSF vs plasma :

Higher in CSF	Equal to plasma	Lower in CSF
• Cl^-. • mg^-.	• Osmolality. • Bicarbonate.	• Glucose : 2/3rd of plasma glucose. • Proteins.

Note : ↑ protein in CSF indicates infection.

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Applied aspect :

1. CSF leak :

- Marker : β_2 transferrin (Present only in CSF).
- Types :
  - CSF rhinorrhea : Leak through nose.
  - CSF otorrhea : Leak through ear.



CSF

2. Post lumbar puncture headache :

LP
↓
CSF loss
↓
Cushioning effect : Lost
↓
Traction on brain
↓
Headache



Lumbar puncture (LP)

Cerebral Blood Flow (CBF) 00:13:05

GENERAL PROPERTIES :

- Blood supply :
  - From circle of Willis.
  - Grey matter > White matter.
- Function : Nutrition.
- Normal CBF : 750 ml/min (or) 14 % of cardiac output (CO).
- O_2 consumption :
  - 20 % of total body resting O_2 consumption (Susceptible to hypoxia).
  - Areas most susceptible to hypoxia :
    1. Hippocampus.
    2. Thalamus.
    3. Inferior colliculus.
    4. Basal ganglia.

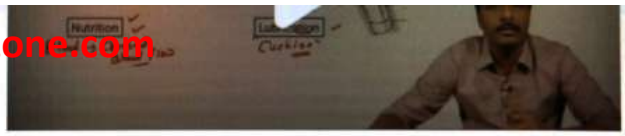
Note : Liver receives highest % of cardiac output.

Factors affecting CBF :

1. Autoregulation :
   - Perfusion pressure range of 65-140 mmHg → CBF constant.
2. Hypercarbia : ↑ pCO2 → ↑ CBF.
3. Hypothermia : ↓ by 1°C → ↓ CBF by 7 %.
   - Significance : Therapeutic hypothermia used in neurosurgeries to ↓ blood loss.

Feedback

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Blood Brain Barrier (BBB)

00:19:40

Structure of BBB :

1. Endothelial cell : Tight junctions.
2. Astrocytes :
   - Star shaped glial cell.
   - Have foot processes.



BBB

Movement across BBB :

Substances able to cross BBB :

1. Lipid soluble substances : Steroid hormones.
2. Gases : O_2, CO_2, nitrogen.
3. Nutrients : Glucose (Via transporters GLUT-1).
4. Water : Via aquaporin 4 channels.

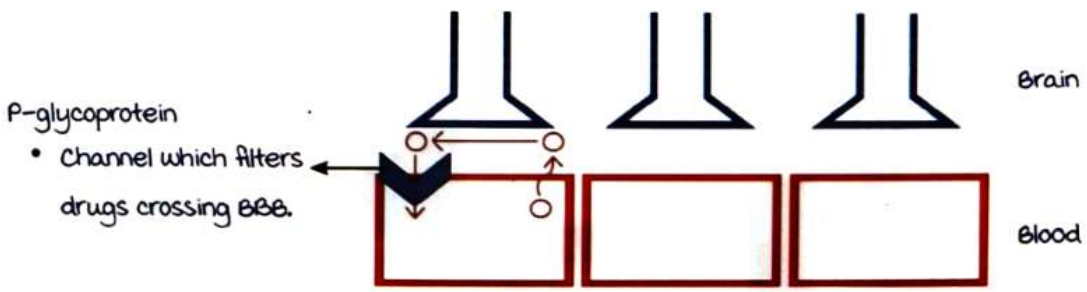
Substances unable to cross BBB :

1. Water soluble substances : Protein hormones.
2. Ions : H^+.
3. Neurotransmitters : Dopamine.

Applied aspect :

- Caissons disease : ↑ levels of N_2 in brain.
- Neuromyelitis optica (Devic's disease) : Autoantibodies produced against aquaporin 4 → Demyelination.
- L-Dopa can cross BBB → used in treatment of parkinsons.

Gatekeepers of BBB :



Significance :

- P-glycoprotein blockers → Concentrate drug in brain.
  - Used in Rx of brain neoplasms.

Feedback

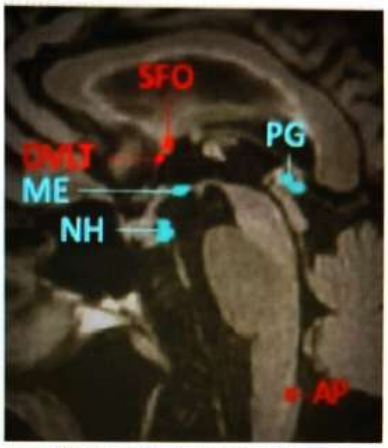


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Circum ventricular organs :

- BBB is absent.
- Mnemonic : SOAP.
  - Sub fornicial organ (SFO).
  - Organ vasculosum of laminae terminalis (OVLT).
  - Area postrema (AP).
  - Posterior pituitary.



Circumventricular organs

Significance :

- Allows passage of protein hormones.
- Eg : Angiotensin II acts on
  - SFO → ↑ thirst.
  - AP → ↑ BP.

Munro Kellie doctrine :

Volume of CSF + Volume of CBF + Volume of brain tissue = Constant.

- Rise in any one of these compresses the other 2.

Eg : CSF blood flow → ↓ blood flow
CSF blood flow → Compresses brain tissue

Feedback